

Contractible Digraphs

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Abstract

Homomorphism graphs are graphs whose vertices are homomorphisms. With our emphasis on homomorphism graphs, in this paper we introduce a non-recursive definition of “dismantlability” for (reflexive or not) digraphs. We show that this definition extends and unifies various definitions of dismantlable structures such as dismantlable graphs, dismantlable digraphs, and dismantlable posets.

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1 Preliminaries

The notion of dismantlable graphs, introduced by T. Poston [3] in his well cited PhD thesis “Fuzzy geometry”, has been proven to be a very useful and interesting concept in the study of graph theory, particularly in connection with the studies of fixed clique properties of graphs and the cop-and-robber games [2, 4, 5].

The work in this paper will revolve around the non-recursive notion of dismantlability for digraphs (which we call *contractible digraphs*, see Definition 1.2). The reason for using the new notion for dismantlable graphs is natural:

- *Extend and unify several known notions of dismantlable structures:* A finite connected poset is dismantlable if it dismantles to a one-point subposet by deleting the irreducible points recursively. The notion of dismantlable posets was introduced by I. Rival [6] for the purpose of characterizing finite posets of length 1 with the fixed point property. In [1], J. Ginsburg has established the following re-

lations of the two notions of dismantlability for posets and graphs:

- a graph is dismantlable if and only if its clique poset is a dismantlable poset; and
- a poset is dismantlable if and only if its comparability graph is a dismantlable graph.

However, at present there are no known extended and unified notion of dismantlability (apart from some new results in [7]) to characterize all (possibly known) dismantlable structures.

We will now set forth the basic definitions and notation to be employed in the sequel: Note that in this paper we concentrate solely on finite digraphs without multiple arcs: A *digraph* $G = (V(G), A(G))$ consists of a nonempty finite set $V(G)$ of elements called *vertices* and a finite set $A(G)$ of ordered pairs of distinct vertices called *arcs*. A vertex x of $V(G)$ is said to be *looped* if $(x, x) \in A(G)$; otherwise x is *loopless*. A digraph G is said to be *reflexive* (resp. *irreflexive*) if all its vertices are looped (resp. loopless). We often consider digraphs which are neither reflexive nor irreflexive.

For digraphs G and H , a homomorphism $f : H \rightarrow G$ maps vertices of H to vertices of G , preserving the arcs, i.e., $(f(x), f(y)) \in A(G)$ whenever $(x, y) \in A(H)$. We denote by $\text{Hom}(H, G)$ the digraph whose vertices are homomorphisms from H to G , with arcs $(f, g) \in A(\text{Hom}(H, G))$ if and only if $(f(x), g(y)) \in A(G)$ whenever $(x, y) \in A(H)$.

Proposition 1.1. *$\text{Hom}(H, G)$ is reflexive for any non-empty digraphs G and H .*

Proof. From the definition of homomorphisms and $\text{Hom}(H, G)$, for any element $f \in V(\text{Hom}(H, G))$, clearly we have $(f(x), f(y)) \in A(G)$ whenever $(x, y) \in A(H)$. Thus $(f, f) \in A(\text{Hom}(H, G))$. $\square$

Recall that a digraph G is connected if its underlying undirected graph is connected.

Definition 1.2. A digraph G is contractible if $\text{Hom}(G, G)$ is connected.

It is easy to check that any null digraph (that is, a digraph consists of a non-empty finite set of vertices and no arcs) is contractible.

Proposition 1.3. Let G be a contractible digraph, and S the induced digraph of G from the set of all looped vertices of G . Then S is connected.

Proof. If the cardinality of S , $\#(S) \leq 1$, then nothing need to be shown. Hence we assume that $\#(S) \geq 2$. Suppose S is not connected. Then we are able to choose two looped vertices $x, y \in V(S) \subseteq V(G)$ such that x and y belong to two different connected components of S . Let H be a reflexive singleton graph. Clearly the constant homomorphisms $x_{H \rightarrow G}$ and $y_{H \rightarrow G}$ are not connected in $\text{Hom}(H, G)$. From Theorem 2.5, $\text{Hom}(G, G)$ is not connected. Thus G is not contractible. Contradiction. $\square$

From Proposition 1.3, the set of looped vertices of a contractible digraph is contained in a connected component of the digraph.

Corollary 1.4. Every reflexive contractible digraph is connected.

Theorem 1.5. Let G be a contractible digraph, and S the induced digraph of G from the set of all looped vertices of G . Then S is contractible.

Proof. From the definition of homomorphisms, each looped vertex of G is mapped to a looped vertex of G by any element of $\text{Hom}(G, G)$, thus it is clear we have $(f_{S \rightarrow S}, g_{S \rightarrow S}) = (f_{G \rightarrow G}|_S, g_{G \rightarrow G}|_S) \in A(\text{Hom}(S, S))$ if $(f_{G \rightarrow G}, g_{G \rightarrow G}) \in A(\text{Hom}(G, G))$. $\square$

2 Homomorphism graphs of contractible digraph

For any self-mapping homomorphism $F : \text{Hom}(H, G) \rightarrow \text{Hom}(H, G)$ and $f \in \text{Hom}(G, G)$, we denote by $f * F$ the homomorphism of $\text{Hom}(H, G)$ to $\text{Hom}(H, G)$ satisfying $(f * F)(g) = f \circ (F(g))$ for each $g \in \text{Hom}(H, G)$.

Lemma 2.1. Let G be a contractible digraph with a looped vertex. Then $\text{Hom}(H, G)$ is contractible for any digraph H .

Proof. Let $x \in V(G)$ be a looped vertex of G . Since $\text{Hom}(G, G)$ is connected, the constant homomorphism $x_{G \rightarrow G}$ is connected to id_G by a path P_x . Let $F, F' : \text{Hom}(H, G) \rightarrow \text{Hom}(H, G)$, be any self-mapping homomorphism. Clearly, $F = \text{id}_G * F$ is connected to $(x_{H \rightarrow G})_{\text{Hom}(H, G) \rightarrow \text{Hom}(H, G)} = x_{G \rightarrow G} * F$ by the path $P_x * F$, where $x_{H \rightarrow G} \in \text{Hom}(H, G)$, $x_{H \rightarrow G}(y) = x$ for all $y \in V(H)$. $\square$

Lemma 2.2. Let G be a contractible digraph with a looped vertex. Then $\text{Hom}(H, G)$ is connected for any digraph H .

Proof. Let $f \in \text{Hom}(H, G)$ be any homomorphism from H to G . Since G is a contractible digraph with a looped vertex, there exists a constant homomorphism x_G connecting to id_G by a path P_x . It is easy to check that the homomorphism $f = \text{id}_G \circ f$ is connected to $x_{H \rightarrow G} = x_G \circ f$ by the path $P_x \circ f$. $\square$

For any self-mapping homomorphism $F : \text{Hom}(G, H) \rightarrow \text{Hom}(G, H)$ and $f \in \text{Hom}(G, G)$, we denote by $F * f$ the homomorphism of $\text{Hom}(G, H)$ to $\text{Hom}(G, H)$ satisfying $(F * f)(g) = F(g \circ f)$ for each $g \in \text{Hom}(G, H)$.

Lemma 2.3. Let G be a contractible digraph with a looped vertex. Then $\text{Hom}(G, H)$ is contractible for any digraph H .

Proof. Let $x \in V(G)$ be a looped vertex of G . Since $\text{Hom}(G, G)$ is connected, the constant homomorphism $x_{G \rightarrow G}$ is connected to id_G by a path P_x . Let $F, F' : \text{Hom}(G, H) \rightarrow \text{Hom}(G, H)$, be any self-mapping homomorphism. Clearly, $F = F * \text{id}_G$ is connected to $F * x_{G \rightarrow G}$ by the path $F * P_x$. $\square$

Lemma 2.4. Let G be a contractible digraph with a looped vertex. Then $\text{Hom}(G, H)$ is connected for any digraph H .

Proof. Let $f \in \text{Hom}(G, H)$ be any homomorphism from G to H . Since G is a contractible digraph with a looped vertex, there exists a constant homomorphism x_G connecting to id_G by a path P_x . Thus it is clear that the homomorphism $f = f \circ \text{id}_G$ is connected to $f(x)_{H \rightarrow G} = f \circ x_G$ by the path $f \circ P_x$. $\square$

Theorem 2.5. Let G be a contractible digraph with a looped vertex. Then for any digraph H , the homomorphism digraphs $\text{Hom}(H, G)$ and $\text{Hom}(G, H)$ are connected and contractible.

Remark 2.6. The conclusion of Lemma 2.2 and 2.4 may be derived straightforwardly from Lemma 2.1, Lemma 2.3, Proposition 1.1, and Corollary 1.4.

3 Contractibility vs dismantlability

In [7], we have extended and unified several known notions of reflexive dismantlable structures such as dismantlable graphs [3] and dismantlable posets [6] by the concept of the *reflexive dismantlable digraphs*. In this section, we shall propose a general version of dismantlable digraphs, and show that the concepts of contractibility and dismantlability are equivalent for general digraphs.

Recall that an induced subgraph A of G is said to be a *retract* of the digraph G if there exists a homomorphism $r : G \rightarrow A$, called *retraction*, such that $r(a) = a$ for all $a \in V(A)$.

Theorem 3.1. *Every retract of a contractible digraph with a looped vertex is contractible.*

Proof. Let A be any retract of the contractible digraph G with $r : G \rightarrow A$ the retraction. It is easy to check that there must exist a looped vertex in A . Since $\text{Hom}(G, G)$ is connected, each constant homomorphism x_G , is connected to id_G by a path P_x . Choose a looped vertex $y \in V(A)$. It is clear that any self-mapping homomorphism $f : A \rightarrow A$ is connected to y_A by the path $r \circ P_x \circ \text{id}_G \circ f$ (note that $f = r \circ \text{id}_G \circ \text{id}_A \circ f$ and $y_A = r \circ y_G \circ \text{id}_A \circ f$). $\square$

For $x \in V(G)$, a vertex y such that $(x, y) \in A(G)$ is a successor of x . A vertex y such that $(y, x) \in A(G)$ is a predecessor of x . The *outset* and the *inset* of a vertex x of G , denoted by $B_G^+(x)$ and $B_G^-(x)$, are the subsets $B_G^+(x) = \{y \mid (x, y) \in A(G)\}$ and $B_G^-(x) = \{y \mid (y, x) \in A(G)\}$, respectively. x is said to be *dominated* by the vertex y of G (or y *dominates* x) if

- (d1) $y \in B_G^+(y) \cap B_G^-(y)$ if $(x, y) \in A(G)$ or $(y, x) \in A(G)$,
- (d2) $B_G^+(x) \setminus \{x, y\} \subseteq B_G^+(y) \setminus \{x, y\}$,
- (d3) $B_G^-(x) \setminus \{x, y\} \subseteq B_G^-(y) \setminus \{x, y\}$.

A finite digraph G is said to be *dismantlable* if there is a linear ordering $<$ on $V(G)$, say $x_0 < \dots < x_n$, such that, for all $i \leq n-1$, there exists $y, y \in \{x_j \in X \mid j > i\}$, such that y dominates x in the induced subgraph $G[\upharpoonright_{< x_i}]$.

Theorem 3.2. *Let G be any finite digraph. Then G is dismantlable if and only if G is contractible.*

Proof. Nothing need to be shown if G is null, otherwise: $(\Rightarrow)$ From the dismantlability of G , it is easy to check that there exists a path P_{x_n} from id_G to x_{nG} in $\text{Hom}(G, G)$. Thus every vertex $f = \text{id}_G \circ f$ of $\text{Hom}(G, G)$ is connected to $x_{nG} = x_{nG} \circ f$. $(\Leftarrow)$ It is clear that id_G is connected to every constant homomorphism x_G . Choose a minimal path, say $\{\text{id}_G, f, \dots, x_G\}$. There is a vertex $y \in V(G)$ satisfying $y \neq f(y) = z$. If $(y, w) \in A(G)$, then $(f(y), \text{id}_G(w)) = (z, w) \in A(G)$; thus $B_G^+(y) \setminus \{y, z\} \subseteq B_G^+(z) \setminus \{y, z\}$. Similarly, $B_G^-(y) \setminus \{y, z\} \subseteq B_G^-(z) \setminus \{y, z\}$ is true. Also it is trivial that $z \in B_G^+(z) \cap B_G^-(z)$ if $(y, z) \in A(G)$ or $(z, y) \in A(G)$. Thus y is dominated by z . Follow Theorem 3.1, we conclude our proof. $\square$

A *partially ordered set (poset)* is a digraph in which arrow relations are reflexive, anti-symmetric and transitive. We adopt conventional notation $\leq_P$ instead of $A(P)$ if P is a poset. Following Rival [6], the point $x \in V(P)$ is said to be an *upper cover* of the point $y \in V(P)$ (or, y is a *lower cover* of x), if, $y <_P x$ and there is no point z of P for which $y <_P z <_P x$. The point z is said to be *irreducible* in P if either z has exactly one upper cover in P or z has exactly one lower cover in P . Finally, the finite poset P is said to be $\leq_P$ -*dismantlable* if $V(P)$ can be linearly ordered as $x_0 < \dots < x_n$, such that, for all $i \leq n-1$, x_i is irreducible in the poset $P \setminus (\downarrow_{< x_i})$.

Theorem 3.3. *Let P be any finite poset. Then P is $\leq_P$ -dismantlable if and only if P is contractible.*

Proof. $(\Rightarrow)$ Suppose that $x \in V(P)$ is irreducible in P , that is, there exists $y \in V(P)$ such that y is the unique upper cover (or lower cover) of x . Then clearly x is dominated by y . $(\Leftarrow)$ If x is dominated by y , then $(\downarrow_{\leq_P} x) \setminus \{x, y\} \subseteq (\downarrow_{\leq_P} y) \setminus \{x, y\}$ and $(\upharpoonright_{\leq_P} x) \setminus \{x, y\} \subseteq (\upharpoonright_{\leq_P} y) \setminus \{x, y\}$. Hence y is the unique upper cover (or lower cover) of x . $\square$

4 Concluding remarks

We have introduced a non-recursive definition of dismantlable digraphs to extend and unify various notions of dismantlable structures. As already mentioned, the dismantlable graphs has been proven to be a very useful and interesting concept in the study of graph theory, particularly in connection with the studies of fixed clique properties of graphs and the cop-and-robber games. In future work we plan to extend the notions of fixed

clique and point properties and cop-robber games, from undirected graphs to digraphs, with special emphasis on dismantlable digraphs.

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